

S4D CLAMMARE

Class									
Student Number									

Marks

Question 15 (4 marks)

A satellite of mass 4000 kg is launched into a circular orbit at an altitude of 250 km above the surface of the Earth.

- (a) Determine the gravitational potential energy of the satellite at the surface of the Earth. 1

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- (b) Determine the gravitational potential energy of the satellite when it is in orbit 250 km above the surface. 1

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- (c) Determine the kinetic energy of the satellite when it is in orbit 250 km above the surface. 2

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Class									
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Marks

Question 16 (6 marks)

Jupiter has a mass of 1.900×10^{27} kg and is orbited by Callisto, one of the four moons of Jupiter that were discovered by Galileo. The table below gives some statistics relating to Callisto:

Callisto Statistics	
Discovered by	Simon Marius & Galileo Galilei
Date of discovery	1610
Mass (kg)	1.08×10^{23}
Equatorial radius (km)	2,400
Orbital radius (km)	1,883,000
Orbital period	?
Escape velocity (km/sec)	2.45

- (a) Calculate the gravitational force exerted on Callisto by Jupiter. 2

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- (b) Calculate the centripetal acceleration of Callisto 2

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- (c) Calculate the orbital period of Callisto. 2

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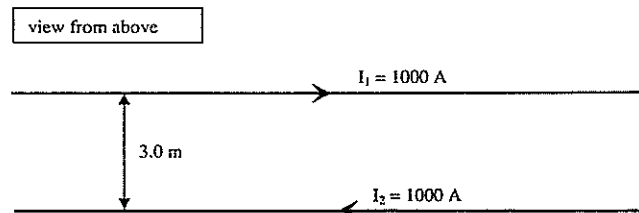
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- 7 A subatomic particle moving at relativistic speed was observed to have a measured mass that was twice its rest mass. What was the velocity of this subatomic particle?

- (A) $v = c$
 (B) $v = \sqrt{3}(c)$
 (C) $v = \left(\frac{\sqrt{3}}{2}\right)(c)$
 (D) $v = \frac{3}{4}(c)$

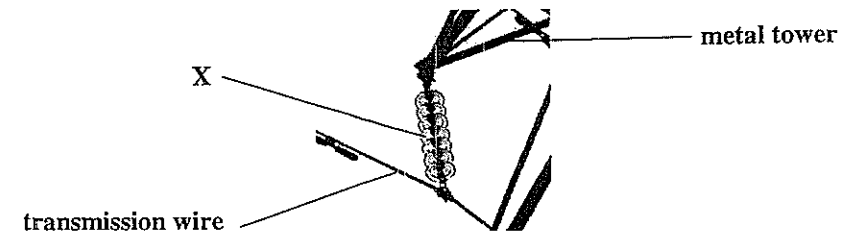
- 8 The electrical power to a small town is supplied by a pair of very long parallel wires, 3.0 metres apart. Each wire carries a current of 1000 A.



What is the magnitude of the electromagnetic force per unit length between the wires?

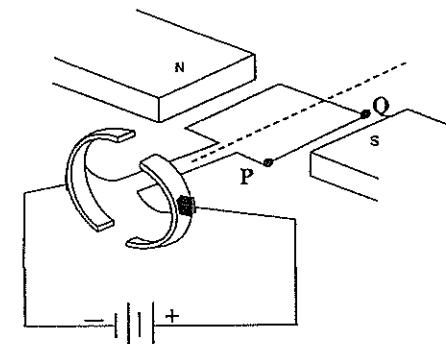
- (A) 0.022 Nm^{-1}
 (B) 0.033 Nm^{-1}
 (C) 0.067 Nm^{-1}
 (D) 0.200 Nm^{-1}

- 9 The structure labelled X in the diagram below is found supporting wires that transport electricity from a power station to a city.



Which of the following statements applies to structure X?

- (A) It is made of ceramic and is superconducting.
 (B) It is made of ceramic and is insulating.
 (C) It is made with a metal core and is conducting.
 (D) It is earthed to protect the transmission wire from lightning strikes.
- 10 The diagram below shows a rectangular coil (single loop) free to rotate in a magnetic field. The length of side PQ of the coil is 5.0 cm and the magnetic field is 0.20 T.



What is the magnitude and direction of electromagnetic force on side PQ of this coil at the position shown if the current is 2.5 A?

- (A) 0.025 N upwards
 (B) 0.025 N downwards
 (C) 2.50 N toward magnetic south.
 (D) 2.50 N downwards

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Question 17 (3 marks)

Sir Isaac Newton used a thought experiment to show that it was theoretically possible to put an object into orbit around the Earth. Describe Newton's thought experiment and include a diagram to illustrate his idea.

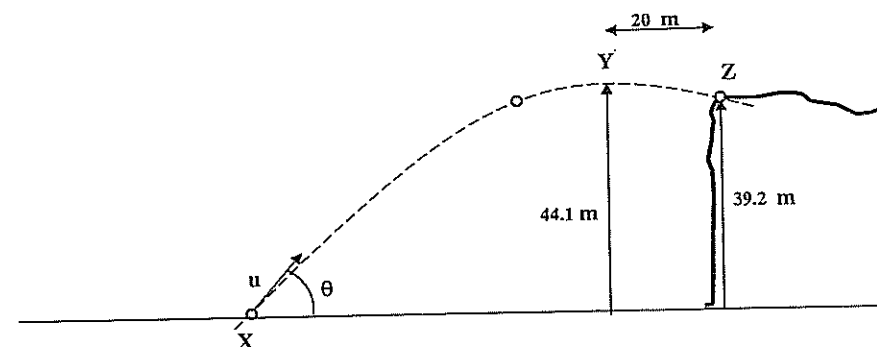
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Class									
Student Number									

Marks

Question 18 (5 marks)

An experimental golf club was used to hit a ball from point X, as illustrated in the following diagram. The ball reaches a maximum height Y, 44.1 metres vertically above the launch point and lands on top of a cliff Z, at a height of 39.2 metres. The horizontal distance from point Y to point Z is 20.0 metres.



- (a) Calculate the time taken for the ball to travel from Y to Z

2

- (b) Determine the magnitude and direction of the initial velocity of projection, u , at point X.

3

Class									
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Marks

Question 19 (3 marks)

An astronaut travelling at $0.89c$ relative to the Earth sets off on a trip to a star. According to the astronaut, the trip takes 2.4 years.

- (a) Determine the distance to the star, according to the astronaut. 1

- (b) Determine the time taken for the trip, according to an observer on the Earth. 1

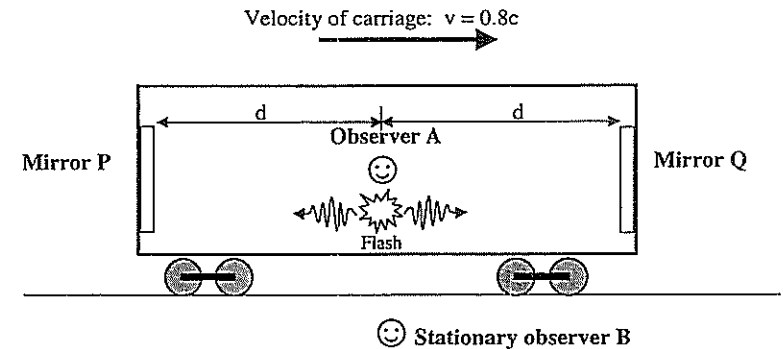
- (c) Determine the distance to the star, according to an observer on the Earth. 1

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Question 20 (4 marks)

In a thought experiment, an observer A, travelling in a train carriage and located at the mid-point of the carriage, sends simultaneous flashes of light to mirrors P and Q located at each end of the carriage as shown in the diagram. The train is moving to the right at a velocity of $0.8c$ and is passing a stationary observer B, on a platform directly opposite A at the moment the flashes are sent.



Analyse this thought experiment to show how it demonstrates the relativity of simultaneity.

4

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Marks

Question 23 (4 marks)

Compare the structure and function of a galvanometer with that of a D.C. electric motor.

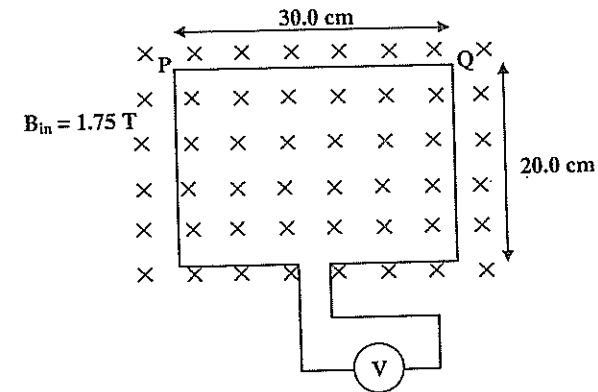
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Marks

Question 24 (4 marks)

The diagram below shows a single rectangular loop of wire of dimensions 20.0 cm x 30.0 cm sitting in a uniform magnetic field of 1.75 T, directed into the page perpendicular to the loop.



- (a) Calculate the magnetic flux through the loop:

2

- (b) (i) The magnetic field strength is reduced to zero at a regular rate over 30 seconds. Calculate the reading on the voltmeter:

1

- (ii) What is direction of the initial induced current between P and Q in the loop as the magnetic field begins to decrease.

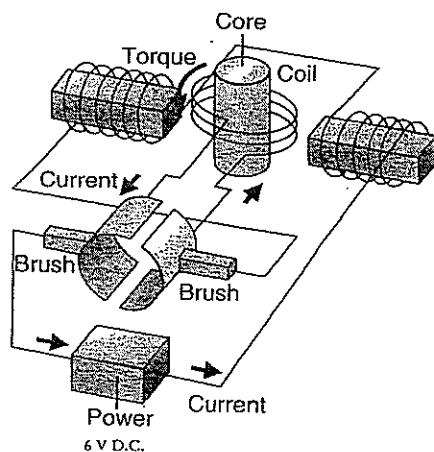
1

Class									
Student Number									

Marks

Question 21 (2 marks)

The diagram below shows the structure of a D.C. motor:



- (a) In the motor show the magnetic field, and identify the method used to produce the magnetic field. 1

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- (b) What causes the motor effect in a D.C. motor? 1

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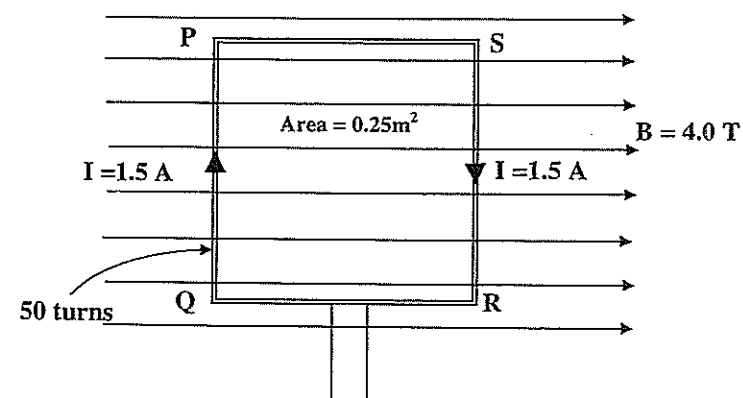
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Marks

Question 22 (3 marks)

The diagram below shows, a square coil of area 0.25 m^2 and consisting of 50 turns of wire, placed in a uniform magnetic field of 4.0 T . A current of 1.5 A flows through the coil in the direction shown:



- (a) Calculate the magnitude and direction of the electromagnetic force on side PQ of the coil. 2

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- (b) Calculate magnitude of the torque on the coil. 1

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Marks

Question 25 (4 marks)

4

In the 1830's Michael Faraday observed that relative motion between a conductor and a bar magnet generates an electric current. Describe an investigation you have performed to investigate the effect on the electric current generated when the relative velocity between the bar magnetic and the conductor is varied.

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Class									
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Marks

Question 26 (3 marks)

Explain why it is relatively easy to turn the shaft of an electrical generator when it is not connected to an external circuit, but much harder when such a connection is made.

3

Question 27 (4 marks)

Explain how an electromagnetic braking system works.

4

Class									
Student Number									

Marks

Question 28 (3 marks)

Outline the competition between Westinghouse and Edison to supply electricity to cities.

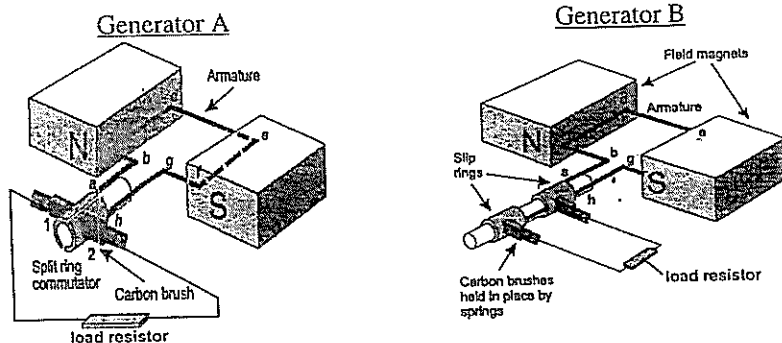
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Class									
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Marks

Question 29 (6 marks)

The following diagram illustrates the structure of two different types of generator.



- (a) Identify each type of generator

2

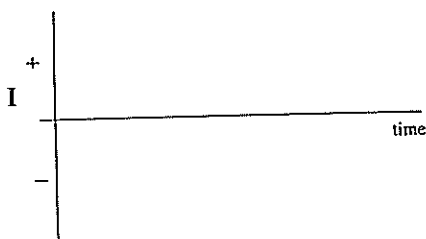
Generator A:

Generator B:

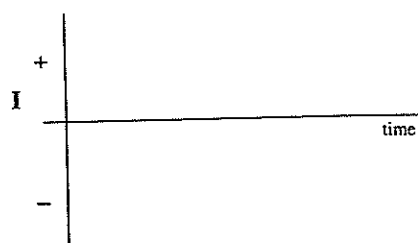
- (b) For each generator in the diagram above, sketch a graph of the output current in the load resistor, for one complete rotation of the coil beginning with the coil in the position shown. (Your choice of positive direction for current in the load resistor is arbitrary)

4

Generator A



Generator B



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Student Number									

Marks

Question 30 (6 marks)

An electric welder for domestic use on 240 V A.C. includes a transformer that has an input current of 10 A from the mains supply. The primary coil has 500 turns and the secondary coil has 25 turns.

- (a) Calculate the output current for this domestic welder.

2

- (b) During the operation of this welder, the output current is much higher than the current input by the mains supply. Explain this difference in terms of conservation of energy.

3

- (c) Explain why a step-up transformer is used for this welder rather than a step-down transformer.

1

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Question 31 (6 marks)

Discuss the need for transformers in the transfer of electrical energy from a power station to its point of use.

6

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FORMULAE SHEET

$$v = f\lambda$$

$$I \propto \frac{1}{d^2}$$

$$\frac{v_1}{v_2} = \frac{\sin i}{\sin r}$$

$$E = \frac{F}{q}$$

$$R = \frac{V}{I}$$

$$P = VI$$

$$\text{Energy} = VIt$$

$$v_{av} = \frac{\Delta r}{\Delta t}$$

$$a_{av} = \frac{\Delta v}{\Delta t} = \frac{v - u}{t}$$

$$\Sigma F = ma$$

$$F = \frac{mv^2}{r}$$

$$E_k = \frac{1}{2}mv^2$$

$$W = Fs$$

$$p = mv$$

$$\text{Impulse} = Ft$$

$$E_p = -\frac{Gm_1m_2}{r}$$

$$F = mg$$

$$v_x^2 = u_x^2$$

$$v = u + at$$

$$v_y^2 = u_y^2 + 2a_y\Delta y$$

$$\Delta x = u_x t$$

$$\Delta y = u_y t + \frac{1}{2}a_y t^2$$

$$\frac{r^3}{T^2} = \frac{GM}{4\pi^2}$$

$$F = \frac{Gm_1m_2}{d^2}$$

$$E = mc^2$$

$$l_v = l_0 \sqrt{1 - \frac{v^2}{c^2}}$$

$$t_v = \frac{t_0}{\sqrt{1 - \frac{v^2}{c^2}}}$$

$$m_v = \frac{m_0}{\sqrt{1 - \frac{v^2}{c^2}}}$$

FORMULAE SHEET

$$\frac{F}{l} = k \frac{I_1 I_2}{d}$$

$$d = \frac{1}{p}$$

$$F = BIl \sin \theta$$

$$M = m - 5 \log \left(\frac{d}{10} \right)$$

$$\tau = Fd$$

$$\frac{I_A}{I_B} = 100(m_B - m_A)/5$$

$$\tau = nBI A \cos \theta$$

$$\frac{V_p}{V_s} = \frac{n_p}{n_s}$$

$$m_1 + m_2 = \frac{4\pi^2 r^3}{GT^3}$$

$$F = qvB \sin \theta$$

$$\frac{1}{\lambda} = R_H \left[\frac{1}{n_f^2} - \frac{1}{n_i^2} \right]$$

$$E = \frac{V}{d}$$

$$\lambda = \frac{h}{mv}$$

$$E = hf$$

$$A_0 = \frac{V_{out}}{V_{in}}$$

$$c = f\lambda$$

$$\frac{V_{out}}{V_{in}} = -\frac{R_f}{R_i}$$

$$Z = \rho v$$

$$\frac{I_r}{I_0} = \frac{[Z_2 - Z_1]^2}{[Z_2 + Z_1]^2}$$

Physics

Data Sheet

Charge on the electron, q_e	$-1.602 \times 10^{-19} \text{ C}$
Mass of electron, m_e	$9.109 \times 10^{-31} \text{ kg}$
Mass of neutron, m_n	$1.675 \times 10^{-27} \text{ kg}$
Mass of proton, m_p	$1.673 \times 10^{-27} \text{ kg}$
Speed of sound in air	340 m s^{-1}
Earth's gravitational acceleration, g	9.8 m s^{-2}
Radius of Earth, R_E	$6.4 \times 10^6 \text{ m}$
Speed of light, c	$3.00 \times 10^8 \text{ m s}^{-1}$
Magnetic force constant, $\left(k \equiv \frac{\mu_0}{2\pi}\right)$	$2 \times 10^{-7} \text{ N A}^{-2}$
Universal gravitational constant, G	$6.7 \times 10^{-11} \text{ N m}^2 \text{ kg}^{-2}$
Mass of Earth	$6.0 \times 10^{24} \text{ kg}$
Planck's constant, h	$6.626 \times 10^{-34} \text{ J s}$
Rydberg's constant, R (hydrogen)	$1.097 \times 10^7 \text{ m}^{-1}$
Atomic mass unit, u	$1.661 \times 10^{-27} \text{ kg}$ 931.5 MeV/c^2
1 eV	$1.602 \times 10^{-19} \text{ J}$
Density of water, ρ	$1.00 \times 10^3 \text{ kg m}^{-3}$
Specific heat capacity of water	$4.18 \times 10^3 \text{ J kg}^{-1} \text{ K}^{-1}$

PERIODIC TABLE OF THE ELEMENTS

KEY		Atomic Number	Symbol of Element	Atomic Weight	Name of Element
		79	Au	197.0	Gold
1	H	1.008			
2	He	4.003			
3	Li	6.941			
4	Be	9.012			
5	B	10.81			
6	C	12.01			
7	N	14.01			
8	O	16.00			
9	F	19.00			
10	Ne	20.18			
11	Na	22.99			
12	Mg	24.31			
13	Al	26.98			
14	Si	28.09			
15	P	30.97			
16	S	32.07			
17	Cl	35.45			
18	Ar	39.95			
19	K	39.10			
20	Ca	40.08			
21	Sc	44.96			
22	Ti	47.87			
23	V	50.94			
24	Cr	52.00			
25	Mn	54.94			
26	Fe	55.85			
27	Co	58.93			
28	Ni	58.69			
29	Cu	63.55			
30	Zn	65.39			
31	Ga	69.72			
32	Ge	72.61			
33	As	74.92			
34	Se	78.96			
35	Br	79.90			
36	Kr	83.80			
37	Rb	85.47			
38	Sr	87.62			
39	Y	88.91			
40	Zr	91.22			
41	Nb	92.91			
42	Mo	95.94			
43	Tc	98.91			
44	Ru	101.1			
45	Rh	102.9			
46	Pd	106.4			
47	Ag	107.9			
48	Cd	112.4			
49	In	114.8			
50	Sn	118.7			
51	Sb	121.8			
52	Te	127.6			
53	I	126.9			
54	Xe	131.3			
55	Cs	132.9			
56	Ba	137.3			
57-71	Lanthanides				
72	Hf	178.5			
73	Ta	180.9			
74	W	183.8			
75	Re	186.2			
76	Os	190.2			
77	Ir	192.2			
78	Pt	195.1			
79	Au	197.0			
80	Hg	200.6			
81	Tl	204.4			
82	Pb	207.2			
83	Bi	209.0			
84	Po	[210.0]			
85	At	[210.0]			
86	Rn	[222.0]			
87	Fr	[223.0]			
88	Ra	[226.0]			
89-103	Actinides				
104	Rf	[261.1]			
105	Db	[263.1]			
106	Sg	[266.1]			
107	Bh	[264.1]			
108	Hs	[265.1]			
109	Uu	[268]			
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288	Uu	[448]			



2005
FORM VI
MAY EXAMINATION

Physics CRIB

General Instructions

- Working time – 2 hours
- Board-approved calculators may be used
- Write using blue or black pen
- Draw diagrams using pencil
- A Data Sheet, Formula Sheet and Periodic Table are provided at the back of this paper
- Write your student number and class at the top of the Multiple Choice Answer Sheet and each page of Part B
- Hand in your Multiple Choice Sheet and all of Section B in one bundle. (Do not staple together)

EXAMINERS: AWW/ MRW /AAH/ SRW

CHECKLIST	
Each boy should have the following:	
1 Question Paper	
1 Multiple Choice Answer Sheet	

Part A
Total marks (12)
Attempt ALL Questions
Allow about 20 minutes for this Part

1. C
2. C
3. D
4. C
5. C
6. D
7. C
8. C
9. B
10. B
11. B
12. B

Total marks (88)

This paper has two parts, Part A and Part B

Part A

Total marks (12)

- Attempt ALL Questions
- Allow about 20 minutes for this Part

Part B

Total marks (76)

- Attempt ALL Questions
- Allow about 1 hour and 40 minutes for this Part

Class

Student Number

Part B

Total marks (76)

Attempt ALL Questions

Allow about 1 hour and 40 minutes for this Part

Answer the questions in the spaces provided

Show all relevant working in questions involving calculations

Marks

Question 13 (2 marks)

Describe two reasons that can account for variations in the acceleration due to gravity at the Earth's surface

2

Any 2 sensible reasons – 2 x (1)

(But must be CLEAR and DISTINCT)NOT: Effect of other Planets, Moon etc.

Class

Student Number

Marks

Question 14 (4 marks)

The table below contains information related to two satellites, P and Q, orbiting the fictitious planet Rammarg:

Satellite	Orbital Radius (m)	Orbital Period (Earth Days)
P	3.77×10^8	16.7
Q	1.22×10^9	T_Q

- (a) Calculate the orbital period,
- T_Q
- , for the satellite Q.

2

$$T_1^2 / r_1^3 = T_2^2 / r_2^3 \quad T_1 = 97 \text{ days or } 8.4 \times 10^6 \text{ s} - (2)$$

↑ Correct Substitution in here only - (1)

- (b) Calculate the mass of the planet Rammarg.

2

$$r^3/T^2 = G.M/4\pi^2 \longrightarrow M = 1.52 \times 10^{25} \text{ kg} - (2)$$

NB : if $T = 16.7$ days is not converted to seconds then $M = 1.14 \times 10^{35} \text{ kg}$

Class									
Student Number									

Marks

Question 15 (4 marks)

A satellite of mass 4000 kg is launched into a circular orbit at an altitude of 250 km above the surface of the Earth.

- (a) Determine the gravitational potential energy of the satellite at the surface of the Earth. 1

$$GM_{EM}/R_E = -2.5 \times 10^{11} \text{ J} - (1)$$

- (b) Determine the gravitational potential energy of the satellite when it is in orbit 250 km above the surface. 1

$$GM_{EM}/(R_E + h) = -2.4 \times 10^{11} \text{ J} - (1)$$

↑ must convert to m!

- (c) Determine the kinetic energy of the satellite when it is in orbit 250 km above the surface. 2

$$\text{Use } KE = -1/2 GPE \longrightarrow KE = 1.2 \times 10^{11} \text{ J} - (2)$$

$$\text{OR } GMm/r^2 = MV^2/r \longrightarrow V = 7.76 \times 10^3 \text{ m/s} - (1)$$

Class									
Student Number									

Marks

Question 16 (6 marks)

Jupiter has a mass of 1.900×10^{27} kg and is orbited by Callisto, one of the four moons of Jupiter that were discovered by Galileo. The table below gives some statistics relating to Callisto:

Callisto Statistics	
Discovered by	Simon Marius & Galileo Galilei
Date of discovery	1610
Mass (kg)	1.08×10^{23}
Equatorial radius (km)	2,400
Orbital radius (km)	$r_o = 1,883,000$
Orbital period	?
Escape velocity (km/sec)	2.45

- (a) Calculate the gravitational force exerted on Callisto by Jupiter. 2

$$F = G(M_J/r_o^2) \cdot M_2 \longrightarrow F = 3.87 \times 10^{21} \text{ N} - (2)$$

correct substitution only - (1)

NB: r_o in km - (1)

- (b) Calculate the centripetal acceleration of Callisto 2

$$a = F/m = 0.0359 \text{ m/s}^2 - (2) ; \text{ correct substitution} - (1)$$

NB: $a = v^2/r$ with $V = 2.45 \text{ km/s} - (0) !!$

- (c) Calculate the orbital period of Callisto. 2

$$r^3/T^2 = GM/4\pi^2 \longrightarrow T = 1.4 \times 10^6 \text{ s} - (2)$$

Correct substitution only - (1)

NB: if $M = 1.08 \times 10^{23} \text{ kg}$ here $\longrightarrow - (0) !!$

Class									
Student Number									

Marks

Question 17 (3 marks)

Sir Isaac Newton used a thought experiment to show that it was theoretically possible to put an object into orbit around the Earth. Describe Newton's thought experiment and include a diagram to illustrate his idea.

3

For full marks, answer must include :

1. Diagram showing Orbital Motion



2. Concept of Launch from High Point

3. Appreciation that Range increases as Velocity of Launch increases

4. Description of Orbital Motion at sufficient velocity

(-1) for any omission.

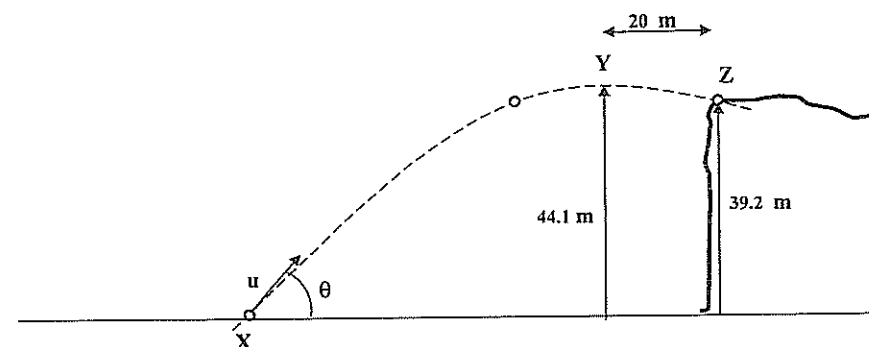
NB : not necessary to discuss "Escape Velocity"

Class									
Student Number									

Marks

Question 18 (5 marks)

An experimental golf club was used to hit a ball from point X, as illustrated in the following diagram. The ball reaches a maximum height Y, 44.1 metres vertically above the launch point and lands on top of a cliff Z, at a height of 39.2 metres. The horizontal distance from point Y to point Z is 20.0 metres.



(a) Calculate the time taken for the ball to travel from Y to Z

2

$$\left. \begin{array}{l} r_y = 4.9\text{m} \\ a = 9.8\text{m/s}^2 \\ u = 0 \end{array} \right\} \begin{array}{l} t = (2r/a)^{1/2} - \text{correct substitution} - (1) \\ t = 1\text{s} - (2) \end{array}$$

(b) Determine the magnitude and direction of the initial velocity of projection, u , at point X.

3

$$\left. \begin{array}{l} u_x = 20\text{m/s from (a)} \\ u_y = 29.4\text{m/s} \end{array} \right\} - (1)$$

$$\uparrow \text{ (from } r = 44.1, u_y = ?, v_y = 0, a_y = -9.8)$$

$$\therefore u = (u_x^2 + u_y^2)^{1/2} \therefore u = 35.6\text{m/s} - (1)$$

$$\tan\theta = u_y/u_x \therefore \theta = 55.8^\circ$$

Class									
Student Number									

Marks

Question 19 (3 marks)

An astronaut travelling at $0.89c$ relative to the Earth sets off on a trip to a star. According to the astronaut, the trip takes 2.4 years.

- (a) Determine the distance to the star, according to the astronaut. 1

$$D = 0.89c \times 2.4 = 2.136 \text{ light years or } 2.02 \times 10^{16} \text{ m}$$

- (b) Determine the time taken for the trip, according to an observer on the Earth. 1

$$T = T_o / \sqrt{1 - v^2/c^2} = 2.4 \times 2.19 = 5.26 \text{ years or } 1.66 \times 10^8 \text{ sec}$$

- (c) Determine the distance to the star, according to an observer on the Earth. 1

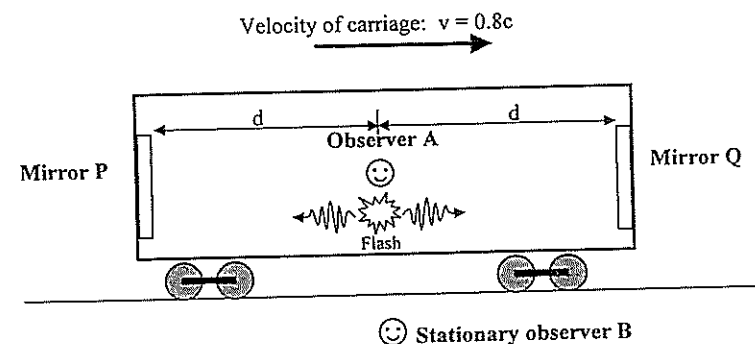
$$D = 5.26 \times 0.89c = 4.68 \text{ light years or } 4.43 \times 10^{16} \text{ m}$$

Class									
Student Number									

Marks

Question 20 (4 marks)

In a thought experiment, an observer A, travelling in a train carriage and located at the mid-point of the carriage, sends simultaneous flashes of light to mirrors P and Q located at each end of the carriage as shown in the diagram. The train is moving to the right at a velocity of $0.8c$ and is passing a stationary observer B, on a platform directly opposite A at the moment the flashes are sent.



Analyse this thought experiment to show how it demonstrates the relativity of simultaneity. 4

Observer A sees both doors open simultaneously as in his reference frame the doors are equally distant and at rest. – (2)

Observer B sees door P open first as P is moving toward the light while Q is moving away from the light. Since the speed of light is the same for both observers they can't agree on the simultaneity of the two events. – (2)

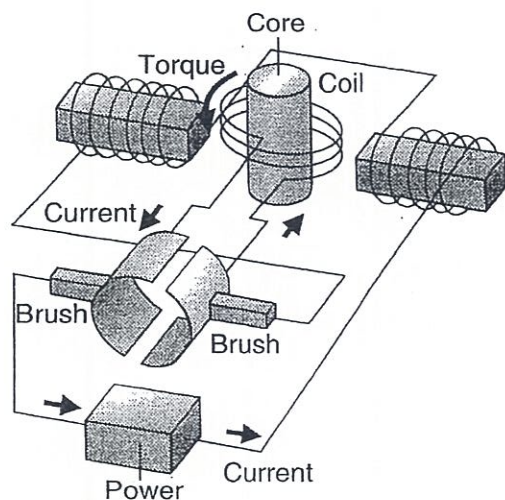
Class

Student Number

Marks

Question 21 (2 marks)

The diagram below shows the structure of a D.C. motor:



- (a) In the motor shown in the diagram, identify the method used to produce the magnetic field. 1

Electromagnet (current carrying coils produce a magnetic field)

- (b) What causes the motor effect in a D.C. motor? 1

Force on a current carrying wire in the magnetic field of the coils.

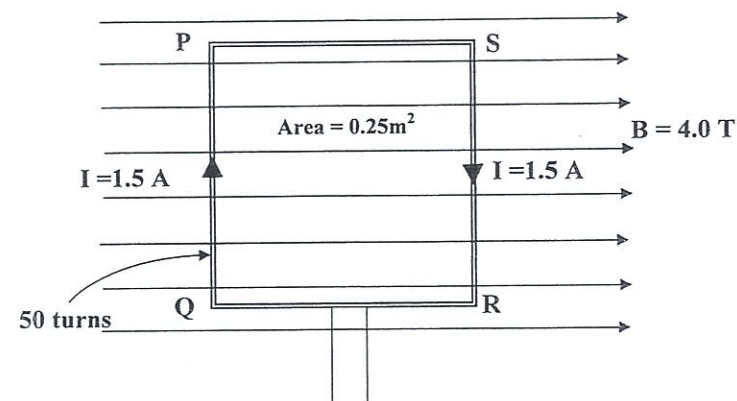
Class

Student Number

Marks

Question 22 (3 marks)

The diagram below shows, a square coil of area 0.25 m^2 and consisting of 50 turns of wire, placed in a uniform magnetic field of 4.0 T . A current of 1.5 A flows through the coil in the direction shown:



- (a) Calculate the magnitude and direction of the electromagnetic force on side PQ of the coil. 2

$$F = nBIL = 150 \text{ N} \text{ (1 mark) into page (1 mark)}$$

NB : downwards is a dangerous expression

- (b) Calculate magnitude of the torque on the coil. 1

$$T = nBIL = 75 \text{ N.m}$$

Class									
Student Number									

Marks

Question 23 (4 marks)

Compare the structure and function of a galvanometer with that of a D.C. electric motor.

4

Marking Scheme.

* Shows an excellent understanding of both motor and galvo with appropriate comparisons of similarities and differences in structure and function. Includes function of both well expressed, motor effect, all parts of spring/commutator, refers to to rotatoin limits/direction and differences. -(4 marks)

* Shows a good understanding of both structure and function with comparison; with some omissions. -(3 marks)

* Describes two aspects of structure and/or function with an attempt to compare but makes critical omissions such as spring or commutator, OR omits motor effect, OR describes one device to the exclusion of the other. -(2 marks)

* Gives at least one correct fact about structure or function. -(1 mark)

Possible Answer: Both have an iron cored rotor coil in a static magnetic field so that when I flows in the coil, the motor effect produces a torque on the coil. In the motor the torque causes a continuous rotation because there is a commutator to reverse the current every half turn however the galvo does not have a commutator but has a restoring spring to stop the rotation when spring torque = motor effect torque on coil; rotation is limited to a max of 90° . The function of the motor is to convert E.E. to kinetic energy whereas the function of the galvo is to measure current.

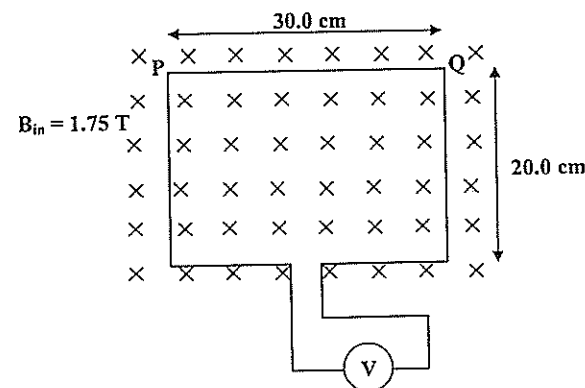
NB : words that imply comparisons are underlined.

Class									
Student Number									

Marks

Question 24 (4 marks)

The diagram below shows a single rectangular loop of wire of dimensions $20.0\text{ cm} \times 30.0\text{ cm}$ sitting in a uniform magnetic field of 1.75 T , directed into the page perpendicular to the loop.



- (a) Calculate the magnetic flux through the loop:

2

$$\phi = BA = 1.75 \times 0.3 \times 0.2 = 0.105 W_b$$

correct answer -(1)

unit W_b -(1)

- (b) (i) The magnetic field strength is reduced to zero at a regular rate over 30 seconds. Calculate the reading on the voltmeter:

1

$$V = \Delta\phi / \Delta t = 0.105/30 = 0.0035V$$

- (ii) What is direction of the initial induced current between P and Q in the loop as the magnetic field begins to decrease.

1

From P to Q, OR right.

Class									
Student Number									
Marks									

Question 25 (4 marks)

In the 1830's Michael Faraday observed that relative motion between a conductor and a bar magnet generates an electric current. Describe an investigation you have performed to investigate the effect on the electric current generated when the relative velocity between the bar magnetic and the conductor is varied.

4

Marking Scheme.

* Hypothesis listed or implied AND/OR lists variables to the tested (velocity and current) - (1 mark)

* lists at least two variables that are kept constant OR describes repetition.

(B field strength, "one magnet"; starting/finishing position of magnet; distance away from magnet/coil) - (1 mark)

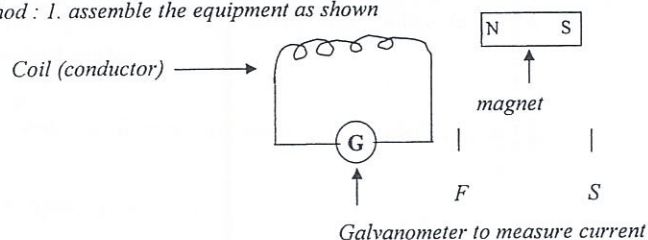
* describes or draws equipment (diagram must be labelled). Need magnet, coil, galvo or ammeter. Voltmeter not accepted. - (1 mark)

* describes result or tabulates expected measurements - (1 mark)

Possible answer:

Aim : To investigate the effect of changing magnet velocity on the induced current in a coil.

Method : 1. assemble the equipment as shown



2. keeping the strength of the magnet and starting and finishing points for movement (S and F) constant move the magnet at high speed toward the coil.

3. Record the maximum current induced.

4. Repeat step 3 and average the current.

5. Repeat steps 2-4 using a low speed.

Expected result : as the speed of the magnet increases the induced current increases.

Class									
Student Number									
Marks									

Question 26 (3 marks)

Explain why it is relatively easy to turn the shaft of an electrical generator when it is not connected to an external circuit, but much harder when such a connection is made.

3

⇒ no connection ∴ no induced current (emf still induced) ∴ no opposing force.

⇒ connection ∴ induced current flows ∴ force produced on induced current which opposes motion ∴ harder to turn.

OR when external circuit connected, current flows ∴ more work needed to produce this energy ∴ more force necessary as $W = Fs$

Note : 1. emf's don't flow

2. extra effort is not due to overcoming "back emf"

3. many boys have only a vague understanding of Lenz's Law – they talked vaguely about an emf or a current induced to oppose the motion (force opposes the motion)

4. do not use the word "resistance" when you mean "force"

5. non connected circuit is ∞ resistance not 0Ω

Question 27 (4 marks)

Explain how an electromagnetic braking system works.

4

Demonstrates a clear, unambiguous understanding of :

1. S = description of physical set-up. i.e. metal disc spinning in a B field.

2. ϕI = motion of conductor in B field ∴ $\Delta \phi$ ∴ emf induced ∴ eddy currents induced.

3. L = force of B on eddy currents opposes motion (Lenz's Law)

4. M = results in a slowing down of the metal disc

Any 3 of the above points - (3 marks)

Any 2 of the above points - (2 marks)

Any 1 of the above points - (1 mark)

Class									
Student Number									
Marks									

Question 28 (3 marks)

Outline the competition between Westinghouse and Edison to supply electricity to cities.

3

Must have: Westinghouse produced AC, Edison produced DC

AND two other relevant, correct features

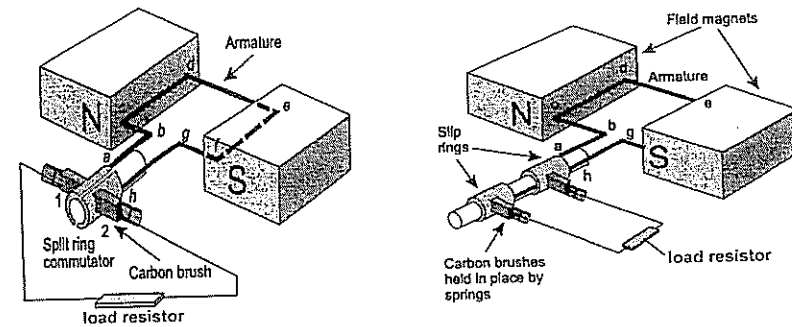
Any two relevant points - (2 marks)

Any one relevant point - (1 mark)

Class									
Student Number									
Marks									

Question 29 (6 marks)

The following diagram illustrates the structure of two different types of generator.



- (a) Identify each type of generator

2

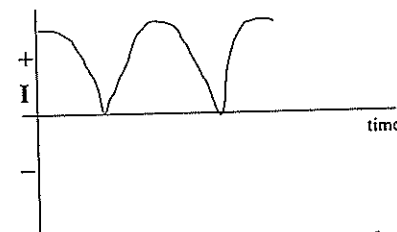
Generator A: *DC generator*

Generator B: *AC generator*

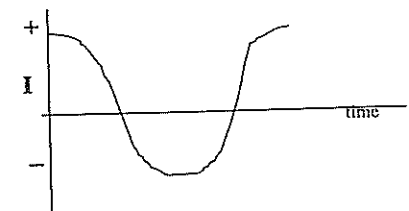
- (b) For each generator in the diagram above, sketch a graph of the output current in the load resistor, for one complete rotation of the coil beginning with the coil in the position shown.
(Your choice of positive direction for current in the load resistor is arbitrary)

4

Generator A



Generator B



For general shape DC - (1 mark); for general shape AC - (1 mark); start at maximum I on both graphs - (1 mark)

Class									
Student Number									

Marks

Question 30 (6 marks)

An electric welder for domestic use on 240 V A.C. includes a transformer that has an input current of 10 A from the mains supply. The primary coil has 500 turns and the secondary coil has 25 turns.

- (a) Calculate the output current for this domestic welder.

$$n_p/n_s = I_s/I_p$$

$$500/25 = I_s/10 \therefore I_s = 200A$$

Incorrect substitution -1 mark

If ratio incorrect, i.e. $n_p/n_s = V_p/V_s = I_p/I_s$ giving 0.5A (1 mark)

- (b) During the operation of this welder, the output current is much higher than the current input by the mains supply. Explain this difference in terms of conservation of energy.

E = statement of conservation of energy of inferred.

$P = \text{conservation of energy requires } E = V_p I_p t = V_s I_s t \text{ hence } V_p I_p = V_s I_s$

I = If stepdown $I_p < I_s$ which means $V_p > V_s$ [when voltages are transformed, the current is inversely affected]

* Any two of the above - (2 marks)

* Any one of the above - (1 mark)

- (c) Explain why a step-up transformer is used for this welder rather than a step-down transformer.

A step down transformer reduces the voltage and increases the current which is used to create a large quantity of heat.

Class _____

Student Number _____

Marks

Question 31 (6 marks)

Discuss the need for transformers in the transfer of electrical energy from a power station to its point of use.

6

[illegible]

Class									
Student Number									

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Physics

Data Sheet

Charge on the electron, q_e	$-1.602 \times 10^{-19} \text{ C}$
Mass of electron, m_e	$9.109 \times 10^{-31} \text{ kg}$
Mass of neutron, m_n	$1.675 \times 10^{-27} \text{ kg}$
Mass of proton, m_p	$1.673 \times 10^{-27} \text{ kg}$
Speed of sound in air	340 m s^{-1}
Earth's gravitational acceleration, g	9.8 m s^{-2}
Radius of Earth, R_E	$6.4 \times 10^6 \text{ m}$
Speed of light, c	$3.00 \times 10^8 \text{ m s}^{-1}$
Magnetic force constant, $\left(k \equiv \frac{\mu_0}{2\pi}\right)$	$2 \times 10^{-7} \text{ N A}^{-2}$
Universal gravitational constant, G	$6.7 \times 10^{-11} \text{ N m}^2 \text{ kg}^{-2}$
Mass of Earth	$6.0 \times 10^{24} \text{ kg}$
Planck's constant, h	$6.626 \times 10^{-34} \text{ J s}$
Rydberg's constant, R (hydrogen)	$1.097 \times 10^7 \text{ m}^{-1}$
Atomic mass unit, u	$1.661 \times 10^{-27} \text{ kg}$ 931.5 MeV/c^2
1 eV	$1.602 \times 10^{-19} \text{ J}$
Density of water, ρ	$1.00 \times 10^3 \text{ kg m}^{-3}$
Specific heat capacity of water	$4.18 \times 10^3 \text{ J kg}^{-1} \text{ K}^{-1}$

FORMULAE SHEET

$$v = f\lambda$$

$$I \propto \frac{1}{d^2}$$

$$\frac{v_1}{v_2} = \frac{\sin i}{\sin r}$$

$$E = \frac{F}{q}$$

$$R = \frac{V}{I}$$

$$P = VI$$

$$\text{Energy} = VIt$$

$$v_{av} = \frac{\Delta x}{\Delta t}$$

$$a_{av} = \frac{\Delta v}{\Delta t} = \frac{v - u}{t}$$

$$\Sigma F = ma$$

$$F = \frac{mv^2}{r}$$

$$E_k = \frac{1}{2}mv^2$$

$$W = Fs$$

$$p = mv$$

$$\text{Impulse} = Ft$$

$$E_p = -\frac{Gm_1m_2}{r}$$

$$F = mg$$

$$v_x^2 = u_x^2$$

$$v = u + at$$

$$v_y^2 = u_y^2 + 2a_y\Delta y$$

$$\Delta x = u_x t$$

$$\Delta y = u_y t + \frac{1}{2}a_y t^2$$

$$\frac{r^3}{T^2} = \frac{GM}{4\pi^2}$$

$$F = \frac{Gm_1m_2}{d^2}$$

$$E = mc^2$$

$$l_v = l_0 \sqrt{1 - \frac{v^2}{c^2}}$$

$$t_v = \frac{t_0}{\sqrt{1 - \frac{v^2}{c^2}}}$$

$$m_v = \frac{m_0}{\sqrt{1 - \frac{v^2}{c^2}}}$$

FORMULAE SHEET

$$\frac{F}{l} = k \frac{I_1 I_2}{d}$$

$$d = \frac{1}{P}$$

$$F = BIl \sin \theta$$

$$M = m - 5 \log \left(\frac{d}{10} \right)$$

$$\tau = Fd$$

$$\frac{I_A}{I_B} = 100(m_B - m_A)/5$$

$$\tau = nBIa \cos \theta$$

$$\frac{V_p}{V_s} = \frac{n_p}{n_s}$$

$$m_1 + m_2 = \frac{4\pi^2 r^3}{GT^2}$$

$$F = qvB \sin \theta$$

$$\frac{1}{\lambda} = R_H \left[\frac{1}{n_f^2} - \frac{1}{n_i^2} \right]$$

$$E = \frac{V}{d}$$

$$\lambda = \frac{h}{mv}$$

$$E = hf$$

$$A_0 = \frac{V_{out}}{V_{in}}$$

$$c = f\lambda$$

$$\frac{V_{out}}{V_{in}} = -\frac{R_f}{R_i}$$

$$Z = \rho v$$

$$\frac{I_r}{I_0} = \frac{[Z_2 - Z_1]^2}{[Z_2 + Z_1]^2}$$

PERIODIC TABLE OF THE ELEMENTS

KEY		Atomic Number		Symbol of element		Name of element	
79	Au	197.0	Gold	79	Au	197.0	Gold
1	H	1.008	Hydrogen	1	H	1.008	Hydrogen
2	He	4.003	Helium	2	He	4.003	Helium
3	Li	6.941	Lithium	3	Li	6.941	Lithium
4	Be	9.012	Beryllium	4	Be	9.012	Beryllium
5	B	10.81	Boron	5	B	10.81	Boron
6	C	12.01	Carbon	6	C	12.01	Carbon
7	N	14.01	Nitrogen	7	N	14.01	Nitrogen
8	O	16.00	Oxygen	8	O	16.00	Oxygen
9	F	19.00	Fluorine	9	F	19.00	Fluorine
10	Ne	20.18	Neon	10	Ne	20.18	Neon
11	Na	22.99	Sodium	11	Na	22.99	Sodium
12	Mg	24.31	Magnesium	12	Mg	24.31	Magnesium
13	Al	26.98	Aluminum	13	Al	26.98	Aluminum
14	Si	28.09	Silicon	14	Si	28.09	Silicon
15	P	30.97	Phosphorus	15	P	30.97	Phosphorus
16	S	32.07	Sulfur	16	S	32.07	Sulfur
17	Cl	35.45	Chlorine	17	Cl	35.45	Chlorine
18	Ar	39.95	Argon	18	Ar	39.95	Argon
19	K	39.10	Potassium	19	K	39.10	Potassium
20	Ca	40.08	Calcium	20	Ca	40.08	Calcium
21	Sc	44.96	Scandium	21	Sc	44.96	Scandium
22	Ti	47.87	Titanium	22	Ti	47.87	Titanium
23	V	50.94	Vanadium	23	V	50.94	Vanadium
24	Cr	52.00	Chromium	24	Cr	52.00	Chromium
25	Mn	54.94	Manganese	25	Mn	54.94	Manganese
26	Fe	55.85	Iron	26	Fe	55.85	Iron
27	Co	58.93	Cobalt	27	Co	58.93	Cobalt
28	Ni	58.69	Nickel	28	Ni	58.69	Nickel
29	Cu	63.55	Copper	29	Cu	63.55	Copper
30	Zn	65.39	Zinc	30	Zn	65.39	Zinc
31	Ga	69.72	Gallium	31	Ga	69.72	Gallium
32	Ge	72.61	Germanium	32	Ge	72.61	Germanium
33	As	74.92	Arsenic	33	As	74.92	Arsenic
34	Se	78.96	Selenium	34	Se	78.96	Selenium
35	Br	79.90	Bromine	35	Br	79.90	Bromine
36	Kr	83.80	Krypton	36	Kr	83.80	Krypton
37	Rb	85.47	Rubidium	37	Rb	85.47	Rubidium
38	Sr	87.62	Strontium	38	Sr	87.62	Strontium
39	Y	88.91	Yttrium	39	Y	88.91	Yttrium
40	Zr	91.22	Zirconium	40	Zr	91.22	Zirconium
41	Nb	92.91	Niobium	41	Nb	92.91	Niobium
42	Mo	95.94	Molybdenum	42	Mo	95.94	Molybdenum
43	Tc	[98.91]	Technetium	43	Tc	[98.91]	Technetium
44	Ru	101.1	Ruthenium	44	Ru	101.1	Ruthenium
45	Rh	102.9	Rhodium	45	Rh	102.9	Rhodium
46	Pd	106.4	Palladium	46	Pd	106.4	Palladium
47	Ag	107.9	Silver	47	Ag	107.9	Silver
48	Cd	112.4	Cadmium	48	Cd	112.4	Cadmium
49	In	114.8	Indium	49	In	114.8	Indium
50	Sn	118.7	Tin	50	Sn	118.7	Tin
51	Sb	121.8	Antimony	51	Sb	121.8	Antimony
52	Te	127.6	Tellurium	52	Te	127.6	Tellurium
53	I	126.9	Iodine	53	I	126.9	Iodine
54	Xe	131.3	Xenon	54	Xe	131.3	Xenon
55	Cs	132.9	Cesium	55	Cs	132.9	Cesium
56	Ba	137.3	Barium	56	Ba	137.3	Barium
57	La	[138.9]	Lanthanum	57	La	[138.9]	Lanthanum
58	Ce	140.1	Cerium	58	Ce	140.1	Cerium
59	Pr	140.9	Praseodymium	59	Pr	140.9	Praseodymium
60	Nd	144.2	Neodymium	60	Nd	144.2	Neodymium
61	Pm	[146.9]	Promethium	61	Pm	[146.9]	Promethium
62	Sm	150.4	Samarium	62	Sm	150.4	Samarium
63	Eu	152.0	Europium	63	Eu	152.0	Europium
64	Gd	157.3	Gadolinium	64	Gd	157.3	Gadolinium
65	Tb	158.9	Terbium	65	Tb	158.9	Terbium
66	Dy	162.5	Dysprosium	66	Dy	162.5	Dysprosium
67	Ho	164.9	Holmium	67	Ho	164.9	Holmium
68	Er	167.3	Erbium	68	Er	167.3	Erbium
69	Tm	168.9	Thulium	69	Tm	168.9	Thulium
70	Yb	173.0	Ytterbium	70	Yb	173.0	Ytterbium
71	Lu	175.0	Lutetium	71	Lu	175.0	Lutetium
72	Hf	178.5	Hafnium	72	Hf	178.5	Hafnium
73	Ta	180.9	Tantalum	73	Ta	180.9	Tantalum
74	W	183.8	Tungsten	74	W	183.8	Tungsten
75	Re	186.2	Rhenium	75	Re	186.2	Rhenium
76	Os	190.2	Osmium	76	Os	190.2	Osmium
77	Ir	192.2	Iridium	77	Ir	192.2	Iridium
78	Pt	195.1	Platinum	78	Pt	195.1	Platinum
79	Au	197.0	Gold	79	Au	197.0	Gold
80	Hg	200.6	Mercury	80	Hg	200.6	Mercury
81	Tl	204.4	Thallium	81	Tl	204.4	Thallium
82	Pb	207.2	Lead	82	Pb	207.2	Lead
83	Bi	209.0	Bismuth	83	Bi	209.0	Bismuth
84	Po	[210.0]	Polonium	84	Po	[210.0]	Polonium
85	At	[210.0]	Astatine	85	At	[210.0]	Astatine
86	Rn	[222.0]	Radon	86	Rn	[222.0]	Radon
87	Fr	[223.0]	Francium	87	Fr	[223.0]	Francium
88	Ra	[226.0]	Radium	88	Ra	[226.0]	Radium
89	Ac	[227.0]	Actinium	89	Ac	[227.0]	Actinium
90	Th	232.0	Thorium	90	Th	232.0	Thorium
91	Pa	231.0	Protactinium	91	Pa	231.0	Protactinium
92	U	238.0	Uranium	92	U	238.0	Uranium
93	Np	[237.0]	Neptunium	93	Np	[237.0]	Neptunium
94	Pu	[239.1]	Plutonium	94	Pu	[239.1]	Plutonium
95	Am	[241.1]	Americium	95	Am	[241.1]	Americium
96	Cm	[244.1]	Curium	96	Cm	[244.1]	Curium
97	Bk	[247.1]	Berkelium	97	Bk	[247.1]	Berkelium
98	Cf	[251.1]	Californium	98	Cf	[251.1]	Californium
99	Es	[252.1]	Einsteinium	99	Es	[252.1]	Einsteinium
100	Fm	[257.1]	Fermium	100	Fm	[257.1]	Fermium
101	Md	[258.1]	Mendelevium	101	Md	[258.1]	Mendelevium
102	No	[259.1]	Nobelium	102	No	[259.1]	Nobelium
103	Lr	[262.1]	Lanthanum	103	Lr	[262.1]	Lanthanum

Lanthanides

57	La	138.9	Lanthanum	57	La	138.9	Lanthanum
58	Ce	140.1	Cerium	58	Ce	140.1	Cerium
59	Pr	140.9	Praseodymium	59	Pr	140.9	Praseodymium
60	Nd	144.2	Neodymium	60	Nd	144.2	Neodymium
61	Pm	[146.9]	Promethium	61	Pm	[146.9]	Promethium
62	Sm	150.4	Samarium	62	Sm	150.4	Samarium
63	Eu	152.0	Europium	63	Eu	152.0	Europium
64	Gd	157.3	Gadolinium	64	Gd	157.3	Gadolinium
65	Tb	158.9	Terbium	65	Tb	158.9	Terbium
66	Dy	162.5	Dysprosium	66	Dy	162.5	Dysprosium
67	Ho	164.9	Holmium	67	Ho	164.9	Holmium
68	Er	167.3	Erbium	68	Er	167.3	Erbium
69	Tm	168.9	Thulium	69	Tm	168.9	Thulium
70	Yb	173.0	Ytterbium	70	Yb	173.0	Ytterbium
71	Lu	175.0	Lutetium	71	Lu	175.0	Lutetium

Actinides

89	Ac	[227.0]	Actinium	89	Ac	[227.0]	Actinium
90	Th	232.0	Thorium	90	Th	232.0	Thorium
91	Pa	231.0	Protactinium	91	Pa	231.0	Protactinium
92	U	238.0	Uranium	92	U	238.0	Uranium
93	Np	[237.0]	Neptunium	93	Np	[237.0]	Neptunium
94	Pu	[239.1]	Plutonium	94	Pu	[239.1]	Plutonium
95	Am	[241.1]	Americium	95	Am	[241.1]	Americium
96	Cm	[244.1]	Curium	96	Cm	[244.1]	Curium
97	Bk	[247.1]	Berkelium	97	Bk	[247.1]	Berkelium
98	Cf	[251.1]	Californium	98	Cf	[251.1]	Californium
99	Es	[252.1]	Einsteinium	99	Es	[252.1]	Einsteinium
100	Fm	[257.1]	Fermium	100	Fm	[257.1]	Fermium
101	Md	[258.1]	Mendelevium	101	Md	[258.1]	Mendelevium
102	No	[259.1]	Nobelium	102	No	[259.1]	Nobelium
103	Lr	[262.1]	Lanthanum	103	Lr	[262.1]	Lanthanum

Where the atomic weight is not known, the relative atomic mass of the most common radioactive isotope is shown in brackets.
The atomic weights of Np and Pu are given for the isotopes ^{237}Np and ^{244}Pu .